

Unit 9

THE PERIODIC TABLE: CHEMICAL PERIODICITY

69 items · 177 marks

Syllabus sections in this unit

- 9.1 Periodicity of physical properties of the elements in Period 3
- 9.2 Periodicity of chemical properties of the elements in Period 3
- 9.3 Chemical periodicity of other elements

Unit 9: The Periodic Table: chemical periodicity

1. M/J 26/P22/Q1(c)

Context

1 Oxygen reacts with many elements.

(c) Li_2O reacts with water to produce a solution with pH greater than 10.

(i) State the name of a Period 3 oxide that behaves in the same way as Li_2O when added to water.

[1]

(ii) Construct an equation for the reaction of Li_2O with water.

[1]

(iii) Identify **all** the Period 3 chlorides that are solid at room temperature and that react with water to produce an acidic solution.

[2]

2. M/J 26/P22/Q1(d)

Context

1 Oxygen reacts with many elements.

(d) LiNO_3 decomposes on heating to produce Li_2O , O_2 and a brown gas.

Identify the brown gas.

[1]

3. M/J 26/P23/Q1(a)

Context

1 The chlorides of Period 3 elements show different physical and chemical properties.

(a) (i) Complete Table 1.1.

Table 1.1

element	formula of chloride	type of bonding in the chloride	type of reaction, if any, of the chloride with cold water
sodium			
silicon			

[3]

(ii) Write an equation for the reaction of PCl_5 with an excess of cold water.

[1]

4. M/J 26/P24/Q3(a)

3 (a) (i) State the trend in atomic radius of the elements across Period 3.

[1]

(ii) State the formula of the oxide formed when sodium reacts with oxygen.

[1]

(iii) Write an equation for the reaction of MgO with water.

[1]

5. M/J 26/P24/Q3(b)

(b) Table 3.1 shows some of the physical properties of Period 3 chlorides.

Table 3.1

	melting point /°C	boiling point /°C	type of bonding	type of structure
NaCl	801	1413		
SiCl₄	−69	58		

(i) Complete Table 3.1 using the abbreviations shown.

M = metallic

C = covalent

I = ionic

S = simple

G = giant

[2]

(ii) State the formula of the phosphorus-containing product formed when PCl_5 reacts with an excess of water. Suggest the pH of the resulting solution.

formula =

pH of solution =

[2]

6. O/N 25/P21/Q2(a)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to Cl.

The ionic radius of Si is **not** shown.

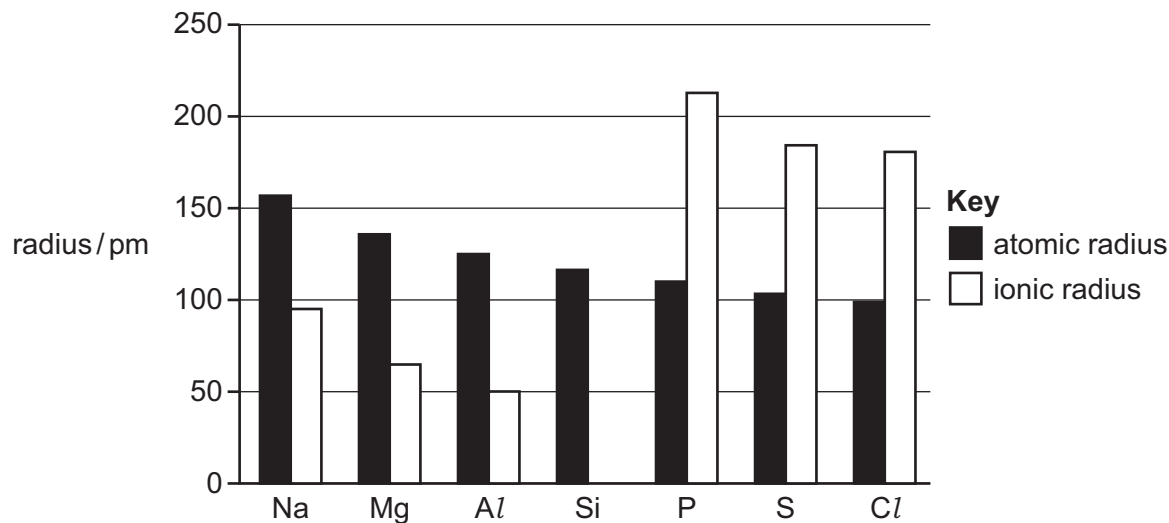


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to Cl.

[2]

(ii) Explain why there is a large difference in the ionic radii of Al and P.

[2]

7. O/N 25/P21/Q2(b)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(b) Table 2.1 gives some information about some of the Period 3 oxides.

Row B gives the pH of the solution that forms when the Period 3 oxide is added to water.

Table 2.1

	formula of Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₃
A	oxidation number of Period 3 element			+3			
B	pH of solution			—	—		

(i) Complete Table 2.1.

[2]

(ii) State why there is no data given in row B for Al₂O₃ and SiO₂.

[1]

8. O/N 25/P21/Q2(c)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(c) (i) Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

[1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form NaAlO_2 .

[1]

9. O/N 25/P22/Q2(a)-(c)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) (i) Explain why the elements Na to Al are good electrical conductors.

[1]

(ii) Explain why the elements P, S and Cl do not conduct electricity.

[1]

(b) Fig. 2.1 shows the variation in melting point of the Period 3 elements Si to Cl.

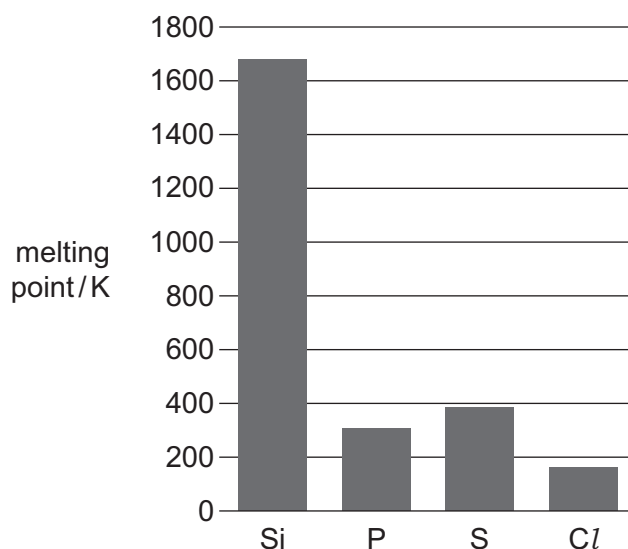


Fig. 2.1

The Period 3 elements Si to Cl are all non-metals.

Explain why there is a large difference between the melting point of Si and the melting points of P, S and Cl.

[2]

(c) Table 2.1 gives some information about some Period 3 chlorides.

Row B refers to the pH of the solution that forms when the Period 3 chloride is added to water.

Table 2.1

	formula of Period 3 chloride	NaCl	MgCl_2	AlCl_3	SiCl_4	PCl_5
A	oxidation number of element bonded to Cl					
B	pH of solution		6.5			
C	bonding			ionic		
D	structure			giant		

Complete Table 2.1.

You may use the following abbreviations.

I = ionic, C = covalent, M = metallic

G = giant, S = simple

[4]

10. O/N 25/P22/Q2(d)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(d) (i) Write an equation for the formation of AlCl_3 from its elements.

[1]

(ii) Write an equation for the formation of H_3PO_4 from PCl_5 .

[1]

11. O/N 25/P23/Q2(a)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to Cl.

The ionic radius of Si is **not** shown.

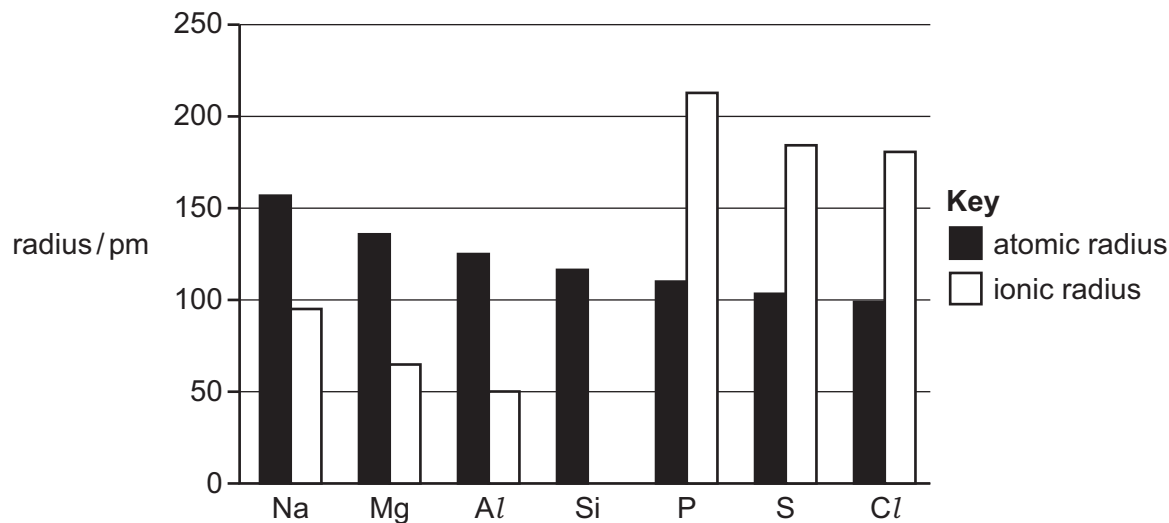


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to Cl.

[2]

(ii) Explain why there is a large difference in the ionic radii of Al and P.

[2]

12. O/N 25/P23/Q2(b)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(b) Table 2.1 gives some information about some of the Period 3 oxides.

Row B gives the pH of the solution that forms when the Period 3 oxide is added to water.

Table 2.1

	formula of Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₃
A	oxidation number of Period 3 element			+3			
B	pH of solution			—	—		

(i) Complete Table 2.1.

[2]

(ii) State why there is no data given in row B for Al₂O₃ and SiO₂.

[1]

13. O/N 25/P23/Q2(c)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(c) (i) Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

[1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form NaAlO_2 .

[1]

14. O/N 25/P24/Q1(a)(ii)

Context

1 The Group 17 elements are oxidising agents.

(ii) Write an equation to show the reaction in which Cl_2 oxidises aluminium metal.

[1]

15. O/N 25/P24/Q1(d)

Context

1 The Group 17 elements are oxidising agents.

(d) Table 1.3 gives some information about MgCl_2 and SiCl_4 .

Table 1.3

	MgCl_2	SiCl_4
electrical conductivity when liquid	conducts	does not conduct
observation when added to water	dissolves	vigorous reaction

(i) Explain the difference between the electrical conductivity of liquid MgCl_2 and of liquid SiCl_4 . Refer to bonding and relevant particles in your answer.

[2]

(ii) Suggest the pH of the solutions that form when each chloride is added to water.

 MgCl_2 SiCl_4

[2]

16. O/N 25/P24/Q2(c)(iii)

Context

- 2** Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

Context: part (b) (introduction)

- (b)** A 5.00 dm^3 sealed flask contains 0.400 mol of $CO(g)$ and 0.800 mol of $H_2(g)$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.

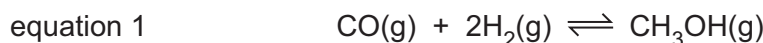


The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

Context: part (c) (introduction)

- (c)** P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH .



P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .

- (iii)** Write an equation to show the reaction of P_4O_{10} with an excess of water.

[1]

17. M/J 25/P21/Q1(b)

- (b)** Separate samples of sodium and sodium oxide are each added to an excess of cold water.

- (i)** Write an equation for the reaction of sodium with cold water.

[1]

- (ii)** Write an equation for the reaction of sodium oxide with cold water.

[1]

- (iii)** Complete Table 1.1.

Do **not** refer to temperature changes when considering observations for these reactions.

Table 1.1

	sodium	sodium oxide
one similarity in observation on addition to cold water		
one difference in observation on addition to cold water		

[2]

18. M/J 25/P21/Q1(c)

- (c) State the oxidation number of the Period 3 elements bonded to Cl in NaCl and PCl_5 .
Explain the difference in the oxidation number.

[2]

19. M/J 25/P22/Q1(b)

- (b) Separate samples of phosphorus(V) chloride and silicon(IV) chloride are each added to an excess of cold water.

- (i) Write an equation for each reaction.

[2]

- (ii) Describe the appearance of phosphorus(V) chloride and silicon(IV) chloride at room temperature.

phosphorus(V) chloride

silicon(IV) chloride

[2]

- (iii) Compare the appearance of the mixtures produced when each reaction is complete.

[2]

20. M/J 25/P22/Q1(c)

- (c) Some oxides are amphoteric.

- (i) Describe what is meant by amphoteric.

[1]

- (ii) Identify the formula of a Period 3 oxide that is amphoteric.

[1]

21. M/J 25/P23/Q2(a)

Context

- 2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

- (a) A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water.
State the formula of the oxide.

[1]

22. M/J 25/P23/Q2(b)

Context

- 2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

- (b) P_4O_{10} is added to an excess of aqueous NaOH.
Write an equation to describe the reaction.

[1]

23. M/J 25/P23/Q2(c)(i)

Context

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

Context: part (c) (introduction)

(c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point/°C
SO ₂	−73
H ₂ O	0
SO ₃	17
SiO ₂	1610
Na ₂ O	1132
MgO	2852
Al ₂ O ₃	2072

- (i)** Identify the oxide from Table 2.1 that contains the element with the highest oxidation number.

[1]

24. M/J 25/P23/Q2(e)

Context

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(e) Al₂O₃ is a white amphoteric compound.

- (i)** State the formula of the aluminium-containing species produced when Al₂O₃ reacts with NaOH(aq).

[1]

- (ii)** State the formula of the aluminium-containing salt produced when Al₂O₃ reacts with H₂SO₄(aq).

[1]

25. M/J 25/P24/Q1(b)

Context

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(b) The relative electrical conductivities of the Period 3 elements are shown in Fig. 1.1.

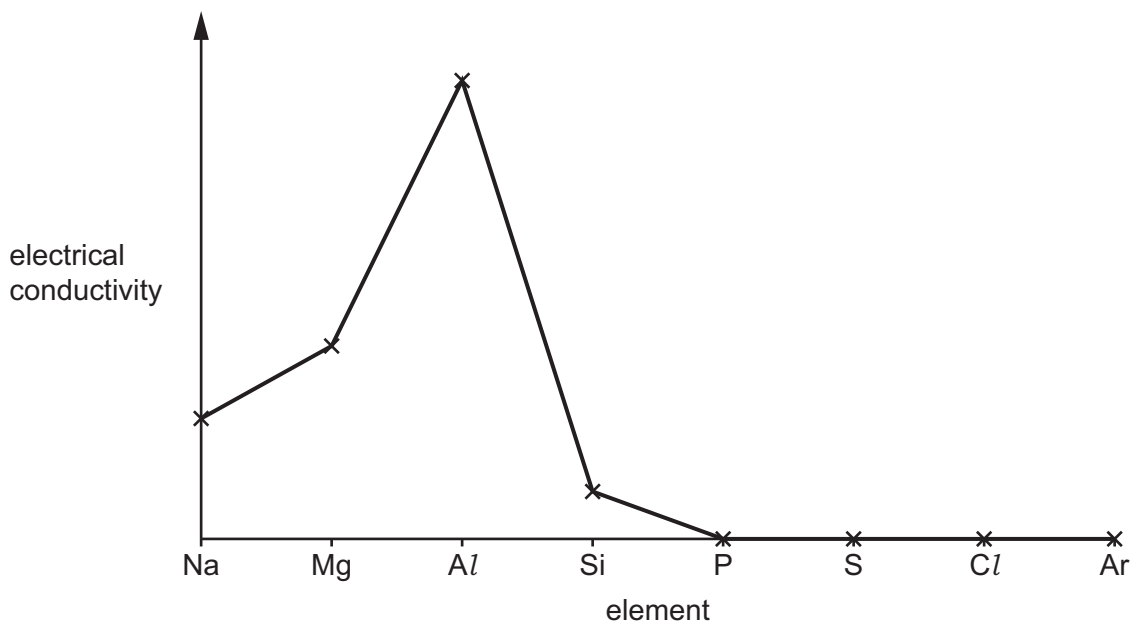


Fig. 1.1

Explain why there is an increase in conductivity from Na to Al and why P, S and Cl are non-conductors of electricity.

[2]

26. M/J 25/P24/Q1(d)

Context

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(d) Most Period 3 elements react with oxygen to form oxides.

(i) Write an equation for the reaction of phosphorus with oxygen.

[1]

(ii) Write an equation for the reaction of aluminium oxide with an excess of aqueous sodium hydroxide.

[1]

27. M/J 25/P24/Q2(b)(i)

Context: part (b) (introduction)

(b) SCl_2 is a red liquid that reacts with water.



(i) Describe **two** observations, other than temperature change, that are made when an excess of water is added to SCl_2 .

[2]

28. M/J 25/P24/Q2(b)(iii)

Context: part (b) (introduction)

(b) SCl_2 is a red liquid that reacts with water.



(iii) State the general name to describe the type of reaction that occurs when a substance is broken down by water.

[1]

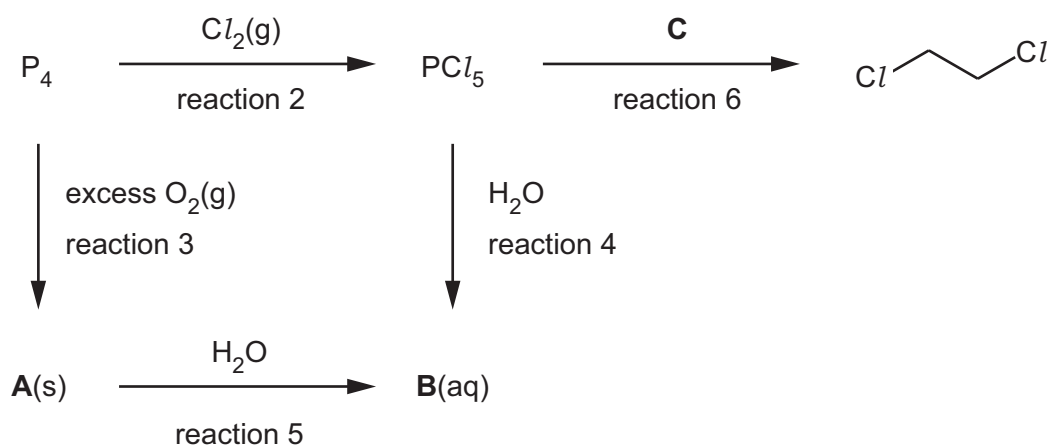
29. MAR 25/P22/Q1(b)(i)

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (b) (introduction)

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .

**Fig. 1.1**

(i) Suggest what is observed in reaction 2.

[1]

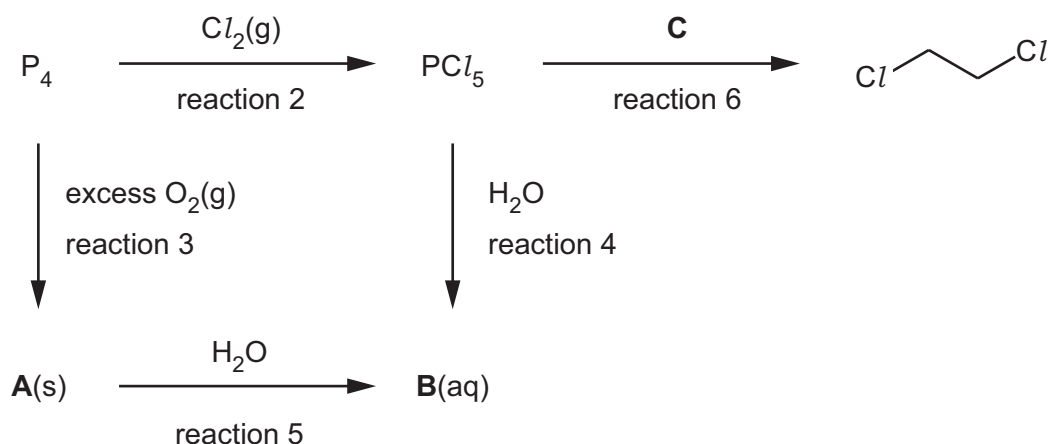
30. MAR 25/P22/Q1(b)(iii)-(v)

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (b) (introduction)

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .

**Fig. 1.1**

(iii) Write an equation for the formation of **A** in reaction 3.

[1]

(iv) Name **B**, which is formed in both reactions 4 and 5.

[1]

(v) Draw the structure of **C**, used in reaction 6.

[1]

31. O/N 24/P21/Q3(e)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (e) (introduction)

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

[1]

(ii) Describe what is observed when a small sample of $SiCl_4$ is added to water.

[1]

32. O/N 24/P21/Q3(f)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(f) Tin forms an amphoteric oxide, SnO_2 .

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

[1]

33. O/N 24/P22/Q2(a)(i)

Context

2 Oxygen is a Group 16 element.**(a) (i)** Write equations for the following reactions.

- sodium and oxygen
- sulfur and oxygen

[2]

34. O/N 24/P22/Q2(a)(iii)

Context

2 Oxygen is a Group 16 element.**(iii)** The maximum oxidation state of the Period 3 elements in their oxides varies across the period.

State and explain the variation.

[2]

35. O/N 24/P22/Q2(b)(i)

Context

2 Oxygen is a Group 16 element.

Context: part (b) (introduction)

(b) H_2O reacts with both inorganic and organic compounds.**(i)** Complete Table 2.1 to give details of the reactions of some Period 3 oxides with H_2O .**Table 2.1**

Period 3 oxide	product of reaction with H_2O	pH of solution formed
	$\text{Mg}(\text{OH})_2$	
P_4O_{10}		

[2]

36. O/N 24/P23/Q3(e)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (e) (introduction)

(e) Silicon readily reacts with elements of high electronegativity.**(i)** Write an equation for the formation of SiCl_4 from its constituent elements.

[1]

(ii) Describe what is observed when a small sample of SiCl_4 is added to water.

[1]

37. O/N 24/P23/Q3(f)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(f) Tin forms an amphoteric oxide, SnO_2 .

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

[1]

38. M/J 24/P21/Q2(a)

2 (a) Sulfur chloride, SCl_2 , is a liquid at room temperature. When SCl_2 is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

(i) Identify the type of reaction that occurs when SCl_2 is added to water.

[1]

(ii) Name a chloride of a different Period 3 element that is also a liquid at room temperature and produces misty fumes when added to water.

[1]

39. M/J 24/P22/Q1(d)-(e)

- (d) Fig. 1.1 shows the variation in melting point of some Period 3 elements in their standard states at room temperature and pressure.

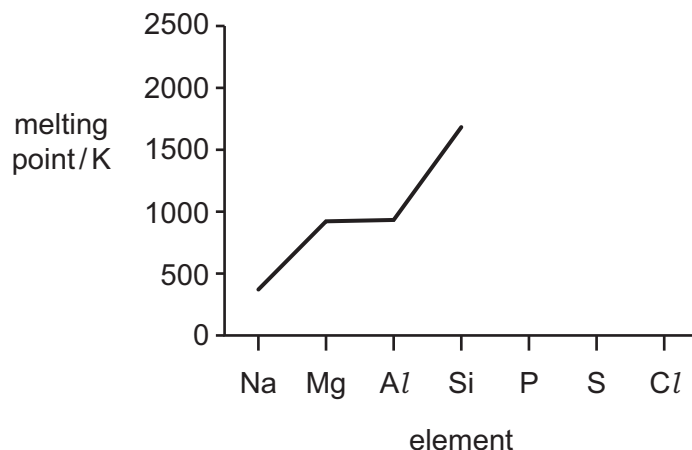


Fig. 1.1
3

- (i) Explain why Si has a high melting point.

[1]

- (ii) Complete Fig. 1.1 to show the variation in the melting points of the elements P, S and Cl.

[2]

- (e) Two Period 3 elements react with an excess of oxygen at room pressure.

- (i) Complete Table 1.2.

Table 1.2

1	2	3
Period 3 element	state of oxide at room temperature and pressure	approximate pH of solution made when oxide is added to water
Na		
S		

[2]

- (ii) The solutions made in column 3 of Table 1.2 are mixed together.
Name the type of reaction that occurs.

[1]

- (iii) Write an equation to describe the reaction between P_4O_{10} and an excess of water.

[1]

40. M/J 24/P22/Q1(f)

(f) Aluminium hydroxide is amphoteric.

(i) Explain what is meant by amphoteric.

[1]

(ii) Write an equation to describe the reaction that occurs when aluminium hydroxide, $\text{Al}(\text{OH})_3$, reacts with $\text{NaOH}(\text{aq})$.

[1]

41. M/J 24/P23/Q1(b)

(b) (i) A sample of Al reacts with an excess of Cl_2 .

State the oxidation number of Al in the product of the reaction.

oxidation number of Al

[1]

(ii) State what determines the maximum oxidation number of the Period 3 elements in their oxides.

[1]

42. M/J 24/P23/Q1(c)

(c) Separate samples of aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , react with an excess of $\text{NaOH}(\text{aq})$ at room temperature.

(i) Give the state of Al_2O_3 and P_4O_{10} at room temperature.

Al_2O_3

P_4O_{10}

[1]

(ii) Write an equation for the reaction of each oxide with an excess of $\text{NaOH}(\text{aq})$ at room temperature.

Al_2O_3 +

P_4O_{10} +

[4]

43. MAR 24/P22/Q2(a)

Context

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

- (a)** Complete Table 2.1.

[3]

44. MAR 24/P22/Q2(c)(ii)

Context

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (c) (introduction)

- (c)** An excess of Cl_2 reacts with phosphorus to form PCl_5 .

- (ii) PCl_5 reacts with **J** to form H_3PO_4 .

Identify **J** and state the type of reaction.

J	type of reaction
---	------------------

[2]

45. O/N 23/P21/Q1(d)(ii)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

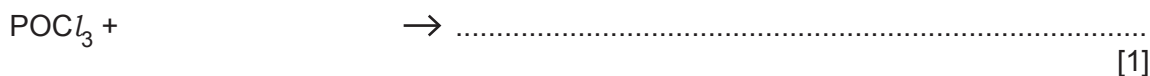
Context: part (d) (introduction)

- (d)** POCl_3 shows similar chemical properties to PCl_5 .

POCl_3 has a melting point of 1°C and a boiling point of 106°C .

POCl_3 reacts vigorously with water, forming misty fumes and an acidic solution.

- (ii) Construct an equation for the reaction of POCl_3 with water.



46. O/N 23/P22/Q1(a)

Context

- 1** The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

- (a) (i)** Describe the variation in atomic radius from silicon to sulfur.

[1]

- (ii) The melting point of silicon is 1410°C . The melting point of sulfur is 113°C .

Explain this difference.

[3]

47. O/N 23/P22/Q3(a)(i)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (a) (introduction)

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4 .

(i) Write an equation for each step.

step 1

step 2

[2]

48. O/N 23/P23/Q1(d)(ii)

Context

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

Context: part (d) (introduction)

(d) POCl_3 shows similar chemical properties to PCl_5 .

POCl_3 has a melting point of 1°C and a boiling point of 106°C .

POCl_3 reacts vigorously with water, forming misty fumes and an acidic solution.

(ii) Construct an equation for the reaction of POCl_3 with water.

$$\text{POCl}_3 + \dots \rightarrow \dots$$

[1]

49. M/J 23/P21/Q1(f)

Context

1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(f) TeF_x reacts with water to form tellurium hydroxide and HF. The oxidation number of tellurium does **not** change during this reaction.

(i) Construct an equation for the reaction of TeF_x with water.

[1]

(ii) Name the type of reaction that occurs when TeF_x reacts with water.

[1]

50. M/J 23/P21/Q3(a)

Context

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance

structure and bonding

[2]

51. M/J 23/P21/Q3(b)

Context

3 Chlorine is a very reactive element.

(b) Samples of magnesium chloride and phosphorus(V) chloride are added to separate beakers of cold water.

Complete Table 3.1. Ignore temperature changes when considering observations for these reactions.

Table 3.1

	magnesium chloride	phosphorus(V) chloride
appearance at room temperature		
one similarity in observation on addition to cold water		
one difference in observation on addition to cold water		
pH of final solution		

[4]

52. M/J 23/P22/Q1(a)(iii)

Context

- 1** The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point /K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (iii) Explain the difference in the melting points of phosphorus and sulfur in terms of structure and bonding.

[2]

53. M/J 23/P22/Q2(a)

Context

- 2** Chlorine is a reactive element. It forms many compounds.

- (a) (i) Complete Table 2.1 to show the maximum oxidation number of the elements Na to P in their chlorides.

Table 2.1

element	Na	Mg	Al	Si	P
maximum oxidation number					

[1]

- (ii) State what determines the maximum oxidation number of elements in Period 3.

[1]

54. M/J 23/P22/Q2(b)

Context

- 2** Chlorine is a reactive element. It forms many compounds.

- (b) An excess of cold water is added to the chloride of silicon.

- (i) Write an equation for the reaction between an excess of cold water and the chloride of silicon.

[1]

- (ii) Suggest the pH of the solution produced in (b)(i).

[1]

55. M/J 23/P22/Q2(c)

Context

2 Chlorine is a reactive element. It forms many compounds.

(c) An excess of cold water is added to the chloride of phosphorus.

(i) Write an equation for the reaction between an excess of cold water and the chloride of phosphorus.

[1]

(ii) Suggest the pH of the solution produced in **(c)(i)**.

[1]

56. M/J 23/P23/Q2(a)

2 (a) The reaction of pure aluminium is only observed if the aluminium oxide layer is removed first. When pure aluminium is added to cold water, bubbles of gas are seen.

(i) State **one** property of aluminium oxide that explains why an aluminium object does **not** react with cold water until the aluminium oxide layer is removed.

[1]

(ii) Write an equation, with state symbols, for the reaction of aluminium oxide with an excess of NaOH(aq).

[2]

(iii) Name **one** other Period 3 element that also produces bubbles of gas when added to cold water.

[1]

57. M/J 23/P23/Q2(c)

(c) Aluminium chloride and phosphorus chloride are both white solids.

- (i) State the maximum oxidation number of aluminium and of phosphorus in these solid chloride salts.

maximum oxidation number of aluminium

maximum oxidation number of phosphorus

[1]

- (ii) State why the maximum oxidation number of aluminium is different from that of phosphorus.

[1]

- (iii) Write an equation for the reaction of solid phosphorus chloride and excess water.

[1]

- (iv) Name the type of reaction that occurs when aluminium chloride is added to water.

[1]

- (v) Explain why the solution produced after aluminium chloride is added to water has a pH of 1–2.

[1]

58. MAR 23/P22/Q1(d)(i)-(ii)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

Context: part (d) (introduction)

- (d) CO₂ and SO₂ are acidic gases.

- (i) Write an equation for the reaction of SO₂ with H₂O.

[1]

- (ii) Write an equation for the reaction of SO₂ with NaOH.

[1]

59. O/N 22/P21/Q3(a)

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



(a) From this list, identify:

(i) all the chlorides that have giant ionic structures in the solid state

[1]

(ii) all the chlorides that react vigorously with water to form strongly acidic solutions

[1]

(iii) the chloride that dissolves in water to form a neutral solution

[1]

(iv) the chloride formed from the **element** with the highest melting point.

[1]

60. O/N 22/P22/Q2(a)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(a) Complete Table 2.1.

- Identify the bonding shown by each chloride under standard conditions.
Use C = covalent, I = ionic, M = metallic.
- Identify the structure shown by each chloride under standard conditions.
Use G = giant, S = simple.
- Deduce the oxidation state of the Period 3 element in each chloride.

[4]

61. O/N 22/P22/Q2(b)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

- (b)** Write equations for the reactions of NaCl and PCl_5 with water. Include state symbols in both equations.

NaCl

PCl_5

[3]

62. O/N 22/P23/Q3(a)

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.

NaCl MgCl_2 AlCl_3 SiCl_4 PCl_5

- (a)** From this list, identify:

(i) all the chlorides that have giant ionic structures in the solid state

[1]

(ii) all the chlorides that react vigorously with water to form strongly acidic solutions

[1]

(iii) the chloride that dissolves in water to form a neutral solution

[1]

(iv) the chloride formed from the **element** with the highest melting point.

[1]

63. M/J 22/P21/Q1(f)(i)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

Context: part (f) (introduction)

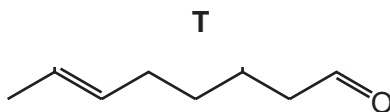
- (f)** Magnesium, Mg, burns in oxygen, O₂.
The activation energy, E_a , for this reaction is +148 kJ mol⁻¹.
- (i)** State **one** observation when magnesium burns in oxygen.
Do **not** refer to temperature changes in your answer.

[1]

64. M/J 22/P21/Q3(d)(iii)

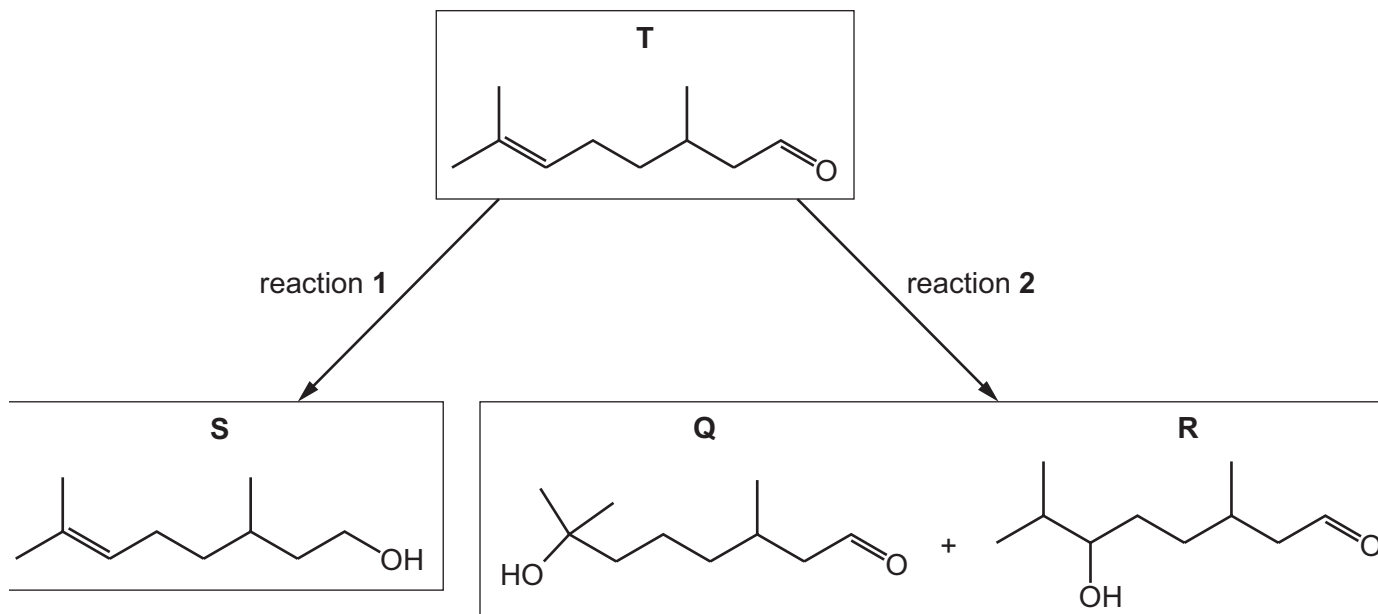
Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

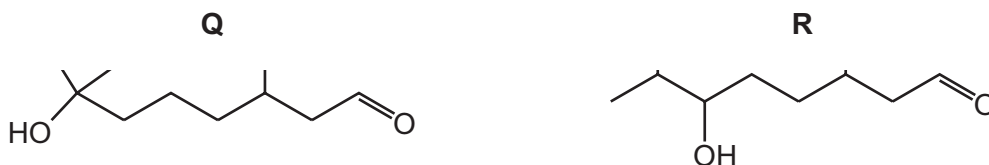
Context: part (c) (introduction)

- (c)** Fig. 3.2 shows two reactions of **T**.

**Fig. 3.2**

Context: part (d) (introduction)

- (d)** Separate samples of **Q** and **R** are added to separate test-tubes containing acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ and heated.

**Fig. 3.3**

- (iii)** Suggest why samples of **Q** and **R** must be dried before PCl_5 is added. Include a relevant equation to support your answer.

65. M/J 22/P23/Q2(a)

2 (a) Period 3 elements and their compounds show trends in their physical properties.

(i) On Fig. 2.1 sketch a graph to show the melting points of the first five elements in Period 3.

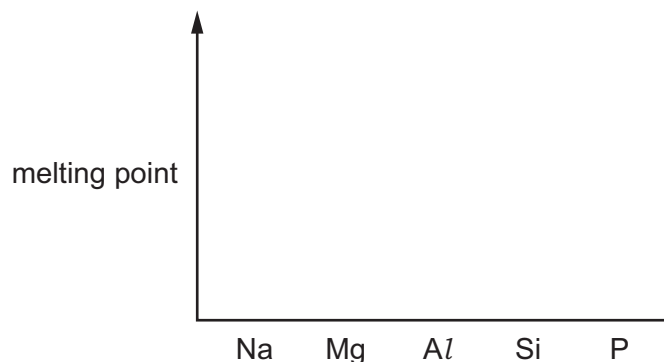


Fig. 2.1

[3]

(ii) Complete Table 2.1 with information for sodium chloride and phosphorus(V) chloride.

Table 2.1

	sodium chloride	phosphorus(V) chloride
state at room temperature		
name of change which occurs on addition of water		
pH of final solution		

[3]

66. MAR 22/P22/Q2(a)(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (a) (introduction)

(a) Na reacts with O_2 to form Na_2O . Na is the reducing agent in this reaction.

(ii) Write an equation for the reaction of Na_2O with water.

[1]

67. MAR 22/P22/Q2(b)(i)-(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (b) (introduction)

(b) Al_2O_3 is an amphoteric oxide found in bauxite.

(i) State what is meant by amphoteric.

[1]

(ii) Al_2O_3 is purified from bauxite in several steps. The first step involves heating Al_2O_3 with an excess of NaOH(aq) . A colourless solution forms.

Write an equation for this reaction.

[1]

68. MAR 22/P22/Q2(c)(iv)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (c) (introduction)

(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(iv) Write an equation for the reaction of P_4O_{10} with water.

[1]

69. MAR 22/P22/Q3(b)(i)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (b) (introduction)

(b) HCl is a product of several different reactions. Some of these are shown in Fig. 3.1.

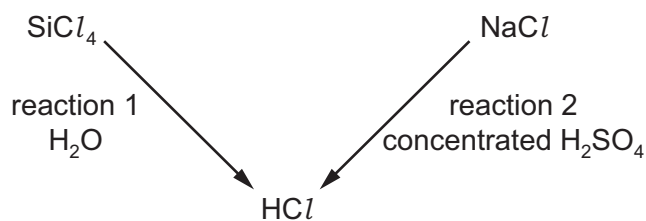


Fig. 3.1

(i) Write an equation for reaction 1.

[1]

Answers Section

1. M/J 26/P22/Q1(c)

1(c)(i)	sodium oxide	1	I formulas BOD sodium
1(c)(ii)	$\text{Li}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{LiOH}$	1	A $2\text{Li}^+ + 2\text{OH}^-$ for 2LiOH BOD LiHO I equations with water as both a reactant and a product
1(c)(iii)	M1 phosphorus (V) chloride / PCl_5 M2 aluminium chloride / AlCl_3	2	M1 A phosphorus pentachloride M1 I phosphorus chloride / phosphorus trichloride M1 I phosphorus alone M2 A Al_2Cl_6 M2 A aluminium alone / aluminium trichloride / aluminium(III) chloride M2 I incorrect aluminium chloride formula I MgCl_2 / magnesium chloride / magnesium I sulfur chlorides / sulfur R silicon chlorides / silicon R NaCl / Na R any period 3 chloride with additional elements R any chloride not from period 3

2. M/J 26/P22/Q1(d)

1(d)	nitrogen dioxide OR NO_2	1	A nitrogen(IV) oxide I nitrogen oxide alone
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3. M/J 26/P23/Q1(a)

1(a)(i)	<table border="1"> <thead> <tr> <th>element</th><th>formula of chloride</th><th>type of bonding</th><th>type of reaction, if any of the chloride with cold water</th></tr> </thead> <tbody> <tr> <td>sodium</td><td>NaCl</td><td>ionic</td><td>no reaction / none A dissolves / dissociates / dissolution</td></tr> <tr> <td>silicon</td><td>SiCl_4</td><td>covalent</td><td>hydrolysis R hydration</td></tr> </tbody> </table>	element	formula of chloride	type of bonding	type of reaction, if any of the chloride with cold water	sodium	NaCl	ionic	no reaction / none A dissolves / dissociates / dissolution	silicon	SiCl_4	covalent	hydrolysis R hydration	3	•✓•✓•✓ I giant / molecular in 'bonding'
element	formula of chloride	type of bonding	type of reaction, if any of the chloride with cold water												
sodium	NaCl	ionic	no reaction / none A dissolves / dissociates / dissolution												
silicon	SiCl_4	covalent	hydrolysis R hydration												
1(a)(ii)	$\text{PCl}_5 + 4 \text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5 \text{HCl}$	1	I $\text{PCl}_5 + \text{H}_2\text{O} \rightarrow \text{POCl}_3 + 2 \text{HCl}$												

4. M/J 26/P24/Q3(a)

3(a)(i)	decreases (across the period)	1	A smaller / goes down A decreases down the period R decreases down the group
3(a)(ii)	Na_2O	1	A Na_2O_2 and NaO_2 A equation that gives Na_2O as the only product I balancing
3(a)(iii)	$\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2$	1	A $\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg}^{2+} + 2\text{OH}^-$ A in equation =

5. M/J 26/P24/Q3(b)

3(b)(i)		type of bonding	type of structure	2	●✓●✓ A
	NaCl	I (ionic)	G (giant)		
	SiCl₄	C (covalent)	S (simple)		

	type of bonding	type of structure
NaCl	GI	GI
SiCl₄	SC	SC

3(b)(ii)	M1 (formula:) H ₃ PO ₄ M2 (pH of solution:) 0–4	2	M1 A PO(OH) ₃ M1 R H ₃ PO ₃ (as excess water) M1 R HCl on formula answer line M1 R PCl ₅ + 4H ₂ O → H ₃ PO ₄ + 5HCl on formula answer line
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6. O/N 25/P21/Q2(a)

2(a)(i)	M1 nuclear charge increases (across period) AND similar shielding M2 greater / stronger attraction to nucleus	2
2(a)(ii)	M1 Al loses (3) electrons / Al is 3+, and P gains (3) electrons / P is 3– M2 Al ⁽³⁺⁾ has (one) fewer shell (than P ^(3–))	2

7. O/N 25/P21/Q2(b)

2(b)(i)	M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6	2
	M2	12-14	8-10	—	—	0-4	0-4	
2(b)(ii)	they are insoluble							1

8. O/N 25/P21/Q2(c)

2(c)(i)	Na ₂ O + 2HCl → 2NaCl + H ₂ O	1
2(c)(ii)	Al ₂ O ₃ + 2NaOH → 2NaAlO ₂ + H ₂ O	1

9. O/N 25/P22/Q2(a)-(c)

2(a)(i)	delocalised electrons move (through the structure / between layers)						1
2(a)(ii)	P / S / Cl have no charge carriers OR (no ions and) no delocalised electrons OR no mobile / moving ions						1
2(b)	M1 Si has (lots of) strong covalent bonds broken M2 P / S / Cl (only) has weak Van der Waals' (forces) broken						2
2(c)	M1	+1	+2	+3	+4	+5	4
	M2	7	6.5	0–4	0–4	0–4	
	M3	I	I	I	C	C	
	M4	G	G	G	S	S	

10. O/N 25/P22/Q2(d)

2(d)(i)	2Al + 3Cl ₂ → 2AlCl ₃	1
2(d)(ii)	PCl ₅ + 4H ₂ O → H ₃ PO ₄ + 5HCl	1

11. O/N 25/P23/Q2(a)

2(a)(i)	M1 nuclear charge increases (across period) AND similar shielding M2 greater / stronger attraction to nucleus	2
2(a)(ii)	M1 Al loses (3) electrons / Al is 3+, and P gains (3) electrons / P is 3– M2 Al ⁽³⁺⁾ has (one) fewer shell (than P ^(3–))	2

12. O/N 25/P23/Q2(b)

2(b)(i)	<table><tr><td>M1</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td><td>(+)6</td></tr><tr><td>M2</td><td>12-14</td><td>8-10</td><td>—</td><td>—</td><td>0-4</td><td>0-4</td></tr></table>	M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6	M2	12-14	8-10	—	—	0-4	0-4	2
M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6										
M2	12-14	8-10	—	—	0-4	0-4										
2(b)(ii)	they are insoluble	1														

13. O/N 25/P23/Q2(c)

2(c)(i)	$\text{Na}_2\text{O} + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O}$	1
2(c)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$	1

14. O/N 25/P24/Q1(a)(ii)

1(a)(ii)	$2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$	1
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15. O/N 25/P24/Q1(d)

1(d)(i)	M1 MgCl_2 (is ionic and) has mobile ions M2 SiCl_4 is covalent AND no charge carriers OR (no ions and) no delocalised electrons OR no mobile / moving ions	2
1(d)(ii)	M1 MgCl_2 pH = 7 M2 SiCl_4 pH 0 – 4	2

16. O/N 25/P24/Q2(c)(iii)

2(c)(iii)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1
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17. M/J 25/P21/Q1(b)

1(b)(i)	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$	1									
1(b)(ii)	$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$	1									
1(b)(iii)	<table border="1"> <tr> <td></td><td>sodium</td><td>sodium oxide</td></tr> <tr> <td>one similarity</td><td colspan="2">(both) solids disappear OR (both produce) colourless solutions</td></tr> <tr> <td>one difference</td><td>effervescence</td><td>no effervescence</td></tr> </table>		sodium	sodium oxide	one similarity	(both) solids disappear OR (both produce) colourless solutions		one difference	effervescence	no effervescence	2
	sodium	sodium oxide									
one similarity	(both) solids disappear OR (both produce) colourless solutions										
one difference	effervescence	no effervescence									

18. M/J 25/P21/Q1(c)

1(c)	M1 $\text{NaCl} + 1$ AND $\text{PCl}_5 + 5$ M2 Na has only 1 valence (shell) electrons AND P has 5	2
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19. M/J 25/P22/Q1(b)

1(b)(i)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	1
	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1
1(b)(ii)	M1 phosphorus(V) chloride white / yellow solid M2 silicon(IV) chloride colourless liquid	2
1(b)(iii)	M1 (phosphorus(V) chloride) produces colourless solution M2 silicon(IV) chloride produces white suspension / white precipitate (in a colorless liquid / mixture / solution)	2

20. M/J 25/P22/Q1(c)

1(c)(i)	species behaves as acid and base OR species reacts with (both) acid and base	1
1(c)(ii)	Al_2O_3	1

21. M/J 25/P23/Q2(a)

2(a)	Na_2O	1
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22. M/J 25/P23/Q2(b)

2(b)	$\text{P}_4\text{O}_{10} + 12\text{NaOH} \rightarrow 4\text{Na}_3\text{PO}_4 + 6\text{H}_2\text{O}$	1
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23. M/J 25/P23/Q2(c)(i)

2(c)(i)	SO_3	1
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24. M/J 25/P23/Q2(e)

2(e)(i)	$\text{NaAl}(\text{OH})_4$	1
2(e)(ii)	$\text{Al}_2(\text{SO}_4)_3$	1

25. M/J 25/P24/Q1(b)

1(b)	M1 Na to Al have (delocalised) electrons that can move (through the structure / lattice) AND (P / S / Cl) do not (have delocalised electrons / mobile charge carriers) M2 Na to Al increasing number of delocalised electrons	2
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26. M/J 25/P24/Q1(d)

1(d)(i)	$\text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
1(d)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$	1

27. M/J 25/P24/Q2(b)(i)

2(b)(i)	<i>In any order:</i> M1 (solution / liquid) (red) to colourless M2 yellow solid appears	2
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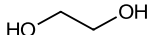
28. M/J 25/P24/Q2(b)(iii)

2(b)(iii)	hydrolysis	1
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29. MAR 25/P22/Q1(b)(i)

1(b)(i)	green colour (of gas) disappears	1
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30. MAR 25/P22/Q1(b)(iii)-(v)

1(b)(iii)	$P_4 + 5O_2 \rightarrow P_4O_{10}$	1
1(b)(iv)	phosphoric(V) acid	1
1(b)(v)		1

31. O/N 24/P21/Q3(e)(i)-(ii)

3(e)(i)	$Si + 2Cl_2 \rightarrow SiCl_4$	1
3(e)(ii)	effervescence / misty / steamy fumes	1

32. O/N 24/P21/Q3(f)

3(f)	$Sn(SO_4)_2$	1
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33. O/N 24/P22/Q2(a)(i)

2(a)(i)	M1 $4Na + O_2 \rightarrow 2Na_2O$	1
	M2 $S + O_2 \rightarrow SO_2$	1

34. O/N 24/P22/Q2(a)(iii)

2(a)(iii)	it increases	1
	as number of valence electrons (which can be used in bonding / lost / shared / donated) increases	1

35. O/N 24/P22/Q2(b)(i)

2(b)(i)	Period 3 oxide	product of rxn with water	pH of solution formed	●✓●✓
	● MgO	Mg(OH) ₂	● 8 ≤ pH ≤ 12	
	P ₄ O ₁₀	● H ₃ PO ₄	● 1 ≤ pH ≤ 4	

36. O/N 24/P23/Q3(e)(i)-(ii)

3(e)(i)	$Si + 2Cl_2 \rightarrow SiCl_4$	1
3(e)(ii)	effervescence / misty / steamy fumes	1

37. O/N 24/P23/Q3(f)

3(f)	$Sn(SO_4)_2$	1
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38. M/J 24/P21/Q2(a)

2(a)(i)	hydrolysis	1
2(a)(ii)	silicon(IV) chloride / silicon tetrachloride ALLOW phosphorus(III) chloride / phosphorus trichloride	1

39. M/J 24/P22/Q1(d)-(e)

1(d)(i)	large amount of energy required to break bonds in giant covalent structure	1						
1(d)(ii)	M1 all 3 melting points lower than Mg M2 show melting point Cl < P < S	2						
1(e)(i)	<table border="1"> <tr> <td>Na</td><td>solid</td><td>10–14</td></tr> <tr> <td>S</td><td>gas</td><td>0–4</td></tr> </table> 4 correct = 2 marks 2 or 3 correct = 1 mark 0 or 1 correct = 0 marks	Na	solid	10–14	S	gas	0–4	2
Na	solid	10–14						
S	gas	0–4						
1(e)(ii)	acid–base OR neutralisation	1						
1(e)(iii)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1						

40. M/J 24/P22/Q1(f)

1(f)(i)	reacts with both acids and bases OR shows both acidic and basic behaviour	1
1(f)(ii)	$\text{Al}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaAl}(\text{OH})_4$	1

41. M/J 24/P23/Q1(b)

1(b)(i)	(+)III	1
1(b)(ii)	number of outer / valence electrons	1

42. M/J 24/P23/Q1(c)

1(c)(i)	both states shown as solid	1
1(c)(ii)	M1 correctly balanced equation for aluminium oxide $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ M2 correctly balanced equation for P_4O_{10} and an excess of NaOH $\text{P}_4\text{O}_{10} + 12\text{NaOH} \rightarrow 4\text{Na}_3\text{PO}_4 + 6\text{H}_2\text{O}$	2

43. MAR 24/P22/Q2(a)

2(a)	formula of chloride	• NaCl	• MgCl ₂	• SiCl ₄
	structure	giant	• giant	• simple
	bonding of chloride	• ionic	• ionic	covalent
	pH of solution formed	• 7	6.2	• 1–4

3

44. MAR 24/P22/Q2(c)(ii)

2(c)(ii)	H ₂ O / water	1
	hydrolysis	1

45. O/N 23/P21/Q1(d)(ii)

1(d)(ii)	$\text{POCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 3\text{HCl}$	1
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46. O/N 23/P22/Q1(a)

1(a)(i)	decreases (from Si to S)	1
1(a)(ii)	M1 Si giant covalent AND S is simple molecular OR Si giant covalent AND S has IMFs	1
	M2 Si involves breaking covalent bonds AND S involves breaking IMFs	1
	M3 more energy needed to break covalent bonds than IMFs OR covalent bonds are strong AND IMFs are weak	1

47. O/N 23/P22/Q3(a)(i)

3(a)(i)	$4\text{P} / \text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1

48. O/N 23/P23/Q1(d)(ii)

1(d)(ii)	$\text{POCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 3\text{HCl}$	1
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49. M/J 23/P21/Q1(f)

1(f)(i)	$\text{TeF}_x + x\text{H}_2\text{O} \rightarrow \text{Te}(\text{OH})_x + x\text{HF}$ OR $\text{TeF}_6 + 6\text{H}_2\text{O} \rightarrow \text{Te}(\text{OH})_6 + 6\text{HF}$	1
1f(ii)	hydrolysis	1

50. M/J 23/P21/Q3(a)

3(a)	M1 <i>appearance</i> (colourless / yellow) liquid M2 <i>structure and bonding</i> simple / molecular AND covalent	2
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51. M/J 23/P21/Q3(b)

3(b)			MgCl_2	PCl_5		4
	appearance		• (white) crystals / solid	• (white OR pale yellow) crystals / solid	[1]	
	one similarity in observation		(colourless) solution made		[1]	
	one difference in observation		no (misty / steamy) fumes	misty / steamy fumes	[1]	
	pH		• 6.5–7	• 0–4	[1]	

52. M/J 23/P22/Q1(a)(iii)

1(a)(iii)	M1 S_8 / molecules of sulfur have more electrons (than P_4 / molecules of phosphorus) M2 S has stronger instantaneous dipole–induced dipole forces (than phosphorus / P)	2
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53. M/J 23/P22/Q2(a)

2(a)(i)		<table><tr><td>element</td><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P</td></tr><tr><td>maximum oxidation number of elements in chlorides</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td></tr></table>	element	Na	Mg	Al	Si	P	maximum oxidation number of elements in chlorides	(+)1	(+)2	(+)3	(+)4	(+)5	All correct for one mark	1
		element	Na	Mg	Al	Si	P									
		maximum oxidation number of elements in chlorides	(+)1	(+)2	(+)3	(+)4	(+)5									
2(a)(ii)	number of outer/valence electrons			1												

54. M/J 23/P22/Q2(b)

2(b)(i)	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1
2(b)(ii)	1–4	1

55. M/J 23/P22/Q2(c)

2(c)(i)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow 5\text{HCl} + \text{H}_3\text{PO}_4$	1
2(c)(ii)	1–4	1

56. M/J 23/P23/Q2(a)

2(a)(i)	it / aluminium oxide is insoluble (in water) OR it / aluminium oxide does not react (with water)	1
2(a)(ii)	M1 correct state of correct aluminium product M2 balanced equation for reaction of aluminium oxide with NaOH $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4(\text{aq})$ OR $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2(\text{aq}) + \text{H}_2\text{O}$	2
2(a)(iii)	sodium (only)	1

57. M/J 23/P23/Q2(c)

2(c)(i)	Al (+)3 / III AND P (+)5 / V	1
2(c)(ii)	different number of valence/outer shell electrons	1
2(c)(iii)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	1
2(c)(iv)	hydrolysis / substitution	1
2(c)(v)	$\text{HCl}(\text{g})$ dissolves in water to make (hydrochloric) acid	1

58. MAR 23/P22/Q1(d)(i)-(ii)

1(d)(i)	$\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$	1
1(d)(ii)	$\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$	1

59. O/N 22/P21/Q3(a)

3(a)(i)	NaCl AND MgCl_2	1
3(a)(ii)	AlCl_3 AND SiCl_4 AND PCl_5	1
3(a)(iii)	NaCl	1
3(a)(iv)	SiCl_4	1

60. O/N 22/P22/Q2(a)

2(a)		NaCl	AlCl ₃	SiCl ₄	PCl ₅	1
		I	I OR C	C	C	
		G	G	S	S	1
		+1	+3	+4	+5	1
				+3	+2	1

61. O/N 22/P22/Q2(b)

2(b)	M1 $\text{NaCl(s)} \rightarrow \text{Na}^{\text{+}}(\text{aq}) + \text{Cl}^{-}(\text{aq})$	1
	$\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl(aq)}$ M2 balanced equation	1
	M3 all state symbols for PCl_5 equation	1

62. O/N 22/P23/Q3(a)

3(a)(i)	NaCl AND MgCl ₂	1
3(a)(ii)	AlCl ₃ AND SiCl ₄ AND PCl ₅	1
3(a)(iii)	NaCl	1
3(a)(iv)	SiCl ₄	1

63. M/J 22/P21/Q1(f)(i)

1(f)(i)	white flame / light OR white solid / smoke	1
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64. M/J 22/P21/Q3(d)(iii)

3(d)(iii)	M1 water reacts with / hydrolyses PCl_5	2
	M2 $\text{H}_2\text{O} + \text{PCl}_5 \rightarrow \text{POCl}_3 + 2\text{HCl}$ OR $4\text{H}_2\text{O} + \text{PCl}_5 \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	

65. M/J 22/P23/Q2(a)

2(a)(i)				3
2(a)(ii)		sodium chloride	phosphorus (V) chloride	3
	state at room temperature	solid	solid	
	name of change which occurs on addition of water	dissolves	hydrolysis	
	pH of final solution	7	1–2	

66. MAR 22/P22/Q2(a)(ii)

2(a)(ii)	$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$	1
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67. MAR 22/P22/Q2(b)(i)-(ii)

2(b)(i)	reacts with both acids and bases / shows both acidic and basic behaviour	1
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2(b)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$	1
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68. MAR 22/P22/Q2(c)(iv)

2(c)(iv)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1
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69. MAR 22/P22/Q3(b)(i)

3(b)(i)	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1
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